

COURSE DESCRIPTION			
Program	Biomedical Engineering, Electronics and Communication, Electronics Engineering, Electronics Engineering (Embedded Systems), Electronics and Computer Engineering, Integrated Circuit Design & Fabrication, Instrumentation Engineering, Micro Electronics		
Course Title	Elements of Electrical & Electronics Engineering Lab		
Course Code	2048	Semester	II
Type of Course	Lab	Contact Hours	45
Teaching Scheme (L: T:P:R)	0:0:3:0	Credits	1.5
CIE Marks	60	ESE Marks	40

Introduction

The Elements of Electrical & Electronics Engineering Lab is a foundational laboratory course designed to provide students with hands-on practical experience in basic electrical and electronics engineering concepts. This course emphasizing experiential learning, skill development, and application of knowledge.

Through structured laboratory experiments, students gain practical exposure to domestic electrical wiring practices, electrical measuring instruments, and basic electrical circuits. The course also enables learners to test and analyze passive and active electronic components, study voltage regulation circuits, and understand the working of amplifiers and digital logic gates.

Course Objectives

1	To familiarize students with domestic electrical wiring materials, standards, and safety practices through hands-on wiring and basic circuit construction experiments.
2	To develop skills in constructing and verifying electrical circuits and in measuring electrical power and energy using standard electrical measuring instruments.
3	To provide practical experience in testing passive and active electronic components and studying the characteristics and operation of Zener diodes and voltage regulator circuits.
4	To enable students to implement digital logic circuits and verify the functional behavior of basic logic gates

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic electrical circuits; passive and active components (concurrent with the semester).	2041	Elements of Electrical & Electronics Engineering	II

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Identify domestic electrical wiring materials and standards, and construct basic single-phase lighting circuits to verify voltage and current division principles.	Apply
CO2	Demonstrate the use of electrical measuring instruments to measure voltage, current, power, and energy in single-phase electrical circuits.	Apply
CO3	Test passive and active electronic components, plot the characteristics of Zener diodes, and demonstrate the operation of fixed and adjustable voltage regulator circuits.	Apply
CO4	Construct basic amplifier and digital logic circuits, and verify the operation of logic gates	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2	2	3	-	-	2	-	-	-
CO2	3	2	-	2	3	1	-	-	-	-	-
CO3	3	2	2	2	3	-	-	-	-	-	-
CO4	3	2	2	2	3	-	-	-	-	-	-
Total	12	8	6	8	12	-	-	2	-	-	-
Correlation Level	3	2	2	2	3	1	-	2	-	-	-

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme					
							Theory			Practical		
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total
0	0	3	0	1.5	45	1.5				60	40	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Detailed Syllabus

Expt No.	Experiment name	Practical Outcome	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
1	Familiarization	Familiarization of Domestic Electrical Wiring Materials and Standards.	CO1	Understand	3
2	Demonstrate domestic Wiring	Demonstrate the wiring of a Single-Phase Domestic Lighting Control Circuit using Source, Fuse, SPST Switch, and Lamp. Demonstrate the wiring of Staircase Lighting Circuit Using Two-Way Switches.	CO1	Apply	3
3	Voltage Division and Current Division Rule	Demonstrate and verify the Voltage Division rule in a Series Circuit containing different power rated Lamps. Demonstrate and verify the Current Division rule in a Parallel Circuit containing different power rated Lamps.	CO1	Apply	3
Module II					
4	Electrical Power Measurement	Demonstrate the usage of Wattmeter to measure electrical power in a single phase circuit.	CO2	Apply	3
5	Electrical Energy Measurement	Demonstrate the usage of an energy meter to measure electrical energy consumption in a single phase electrical circuit.	CO2	Apply	3
6	Extension Board	Wiring and Assembly of an Extension Board with Fuse, Sockets and Switches.	CO2	Apply	3
	Series Exam - I		CO2	Remember	3
Module III					
7	Zener voltage Regulator	Study the line and Load regulation characteristics of Zener voltage regulator	CO3	Apply	3
8	Voltage Regulator ICs	Demonstrate Positive and Negative Voltage Regulation using ICs (78XX & 79XX)	CO3	Apply	3
9	Adjustable Voltage regulator IC	Demonstrate the working of adjustable voltage regulator using LM317. Plot load and line regulation characteristics	CO3	Apply	3
Module IV					
10	Familiarization of Digital trainer kit	Understand the usage of digital trainer kit	CO4	Apply	3
11	Logic functions using discrete components	Demonstrate AND and OR logic functions using discrete components (switches, diodes, resistors etc)	CO4	Apply	3
12	Familiarization of basic Logic Gates	Verify the truth tables of logic gates NOT, AND, OR, NAND, NOR, ExOR, and ExNOR	CO4	Apply	3
	Series Exam - II		CO4	Apply	3
	Open Ended Experiment		CO4	Apply	3

Open-ended experiment

Objective:

To encourage creativity, innovation, and application of knowledge by allowing students to design and perform an experiment beyond the prescribed list, using the concepts learned during the course.

Description:

At the end of the laboratory course, each student (or group of students) shall undertake one open-ended experiment. In this activity, students are expected to conceive, design, and implement an experiment of their own choice related to the Course outcomes. The aim is to apply theoretical knowledge and laboratory skills to a new or practical problem not explicitly covered in the regular experiments. Any 3 to 5 open ended problems may be chosen and divided among student groups.

Students may choose to:

Modify or extend an existing experiment to explore additional parameters or behaviors.

Integrate multiple concepts from the syllabus into a combined application.

Design a mini-setup or prototype to demonstrate a principle or solve a simple engineering problem.

Perform a comparative study or performance analysis of a circuit system/component.

Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Describe why the switch is connected in the phase line. What will happen if the switch is connected in the neutral line instead of the phase line?	1.5
2	Identify various electrical safety mechanisms available in your home.	1.5
3	Explain the relationship between lamp power rating, voltage drop, and brightness in a series circuit	1.5
4	Identify whether domestic electrical appliances are connected in series or parallel. Explain why this arrangement is used.	1.5
Module II		
5	List various electrical appliances used in your home. Identify their power ratings and classify them as high-power and low-power devices.	1.5
6	Identify the energy star rating of an electrical appliance in your home and explain what this rating reveals about its energy efficiency.	1.5
7	Record the initial and final energy meter readings of your home for a chosen time period. Calculate the number of units consumed and estimate the electricity bill for that period. Compare it with the energy bill given by the electricity board.	1.5
Module III		
8	Prepare a chart containing the passive and active electronic components with their diagrams and symbols with proper labeling	1.5
9	Prepare a list of various types of diodes with their key features.	1.5
10	Create a chart illustrating the step-by-step multimeter testing procedures for a diode and a transistor with diagrams.	1.5
11	Prepare a chart containing various fixed voltage regulator ICs with their pinout diagram.	1.5
Module IV		
12	Using any online tool, simulate the working of a NAND gate.	1.5
13	Prepare a document that explains the working of a transistor based inverter logic circuit	1.5
14	Prepare a list of various logic families and write one key feature of each.	1.5
15	Draw the diode-logic implementation of NAND and NOR gates and verify their truth tables	1.5
Total Self-Learning hours		22.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Basic Electronics: A Text-lab Manual	Paul B. Zbar, Albert Paul Malvino, Michael A. Miller	McGraw-Hill Publishing Company

REFERENCE			
Sl. No.	Title	Author	Publication
1	Electronics Lab Manuals	K A Navas	PHI
2	Electronics Lab Manuals	Syammohan S, Kuriachan D	Ayodya

WEB RESOURCES		
Sl. No.	URL	Modules Covered

1	https://www.digimat.in/nptel/courses/video/108105112/L01.html	I & II
2	https://www.digimat.in/nptel/courses/video/108101091/L01.html	III & IV

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Digital Storage Oscilloscope	3 MHz, Dual channel	15
2	Signal/ Function Generator	3 MHz minimum	15
3	Power Supply Unit	30V, Dual	15
4	Multimeter	DMM	15
5	Ammeter	0-2A	15
6	Voltmeter	0-230V	15
7	Energy Meter	230V, 50Hz	15
8	Digital Trainer Kit		15

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology - Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE
		Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Open-ended experiment	Practical
Duration			3 hours	3 hours		3 hours
Exam mark		50	40	40	50	40
Converted to		30	15	15	10	
Marks	5	30	15		10	
Total marks	60					40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test & End Semester Evaluation

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10

Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Scheme of Evaluation - Open-ended experiment

Category	Things to evaluate	Marks
Originality and Relevance of the Idea	Assesses how creative, innovative, and relevant the chosen experiment is to the course objectives.	10
Clarity of Objectives and Experimental Plan	Evaluates how clearly the student defines the aim, scope, and systematic plan of the experiment.	10
Correctness of Execution and Data Recording	Checks the accuracy, safety, and neatness in performing the experiment and recording observations.	10
Analysis, Interpretation, and Presentation of Results	Measures the student's ability to analyze data, interpret outcomes, and present results logically.	10
Teamwork and Innovation Demonstrated	Judges the level of collaboration, initiative, and creativity shown during the experiment.	10
Total		50

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